

Inference for linear regression models

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

What is our population model?

$$\text{Calories} = \beta_0 + \beta_1 \text{ Sugar} + \varepsilon$$

- contains ε
- no hats
- don't know the population parameters, so use β_0, β_1

Activity

Coefficients:

	Estimate	Std. Error
(Intercept)	87.4277	5.1627
Sugar	2.4808	0.7074

How would I interpret the estimated slope?

A one gram increase in amount sugar per serving of breakfast cereal is associated with an increase of 2.48 calories per serving, on average

Activity

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

What are our null and alternative hypotheses?

$$H_0: \beta_1 = 0 \quad (\text{no relationship})$$

$$H_A: \beta_1 > 0 \quad (\text{positive relationship})$$

- in terms of a model parameter (e.g. β_1)
- no hats!

Activity

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 > 0$$

obs estimating
2 parameter
 (β_0, β_1)

Coefficients:

	Estimate	Std. Error
(Intercept)	87.4277	5.1627
Sugar	<u>2.4808</u>	<u>0.7074</u>

$$df = n - 2 \\ = 36 - 2 = 34$$

What is our test statistic? What are the degrees of freedom?

$$t = \frac{\overset{\hat{\beta}_1}{\underset{\leftarrow}{2.4808}} - \underset{\leftarrow \text{value under } H_0}{0}}{\underset{\leftarrow SE \hat{\beta}_1}{0.7074}}$$

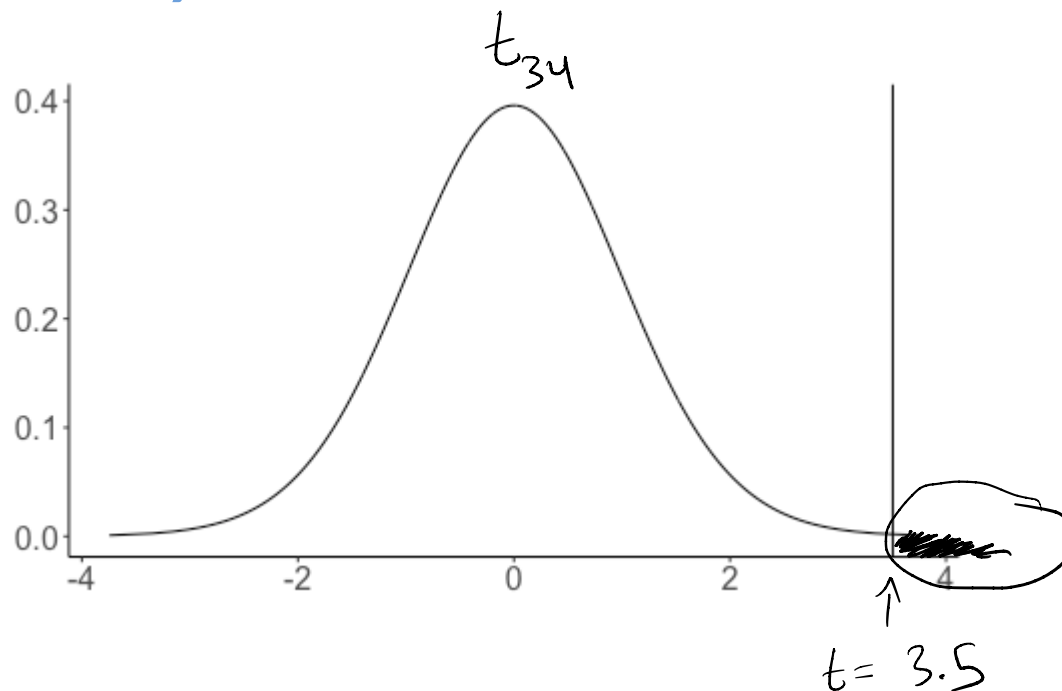
$$\approx 3.5$$

$\hat{\beta}_1$ is 3.5 standard deviations above hypothesized value of 0 from H_0

Activity

$$H_A: \beta_1 > 0$$

$\Rightarrow$ right tail



What area under the curve corresponds to my p-value?

A new question

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

Coefficients:

	Estimate	Std. Error
(Intercept)	87.4277	5.1627
Sugar	2.4808	0.7074

Here $\hat{\beta}_1 = 2.48$. Will $\hat{\beta}_1$ be exactly equal to the true value β_1 ?

A new question

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

Coefficients:

	Estimate	Std. Error
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Here $\hat{\beta}_1 = 2.48$. Will $\hat{\beta}_1$ be exactly equal to the true value β_1 ?

No -- our estimate comes from one sample of data, and there is variability from sample to sample.

How can I account for this variability when reporting an estimate of β_1 ?

Interval estimates

Point estimate: a single "best guess" value of a parameter. The estimated slope $\hat{\beta}_1$ is a point estimate of β_1

Interval estimate: a "reasonable range" of possible values for a parameter. A *confidence interval* is an interval estimate

How would I construct a confidence interval for β_1 ?

$$\hat{\beta}_1 \pm t^* \underbrace{SE \hat{\beta}_1}_{\substack{\uparrow \\ \text{standard error of } \hat{\beta}_1}}$$

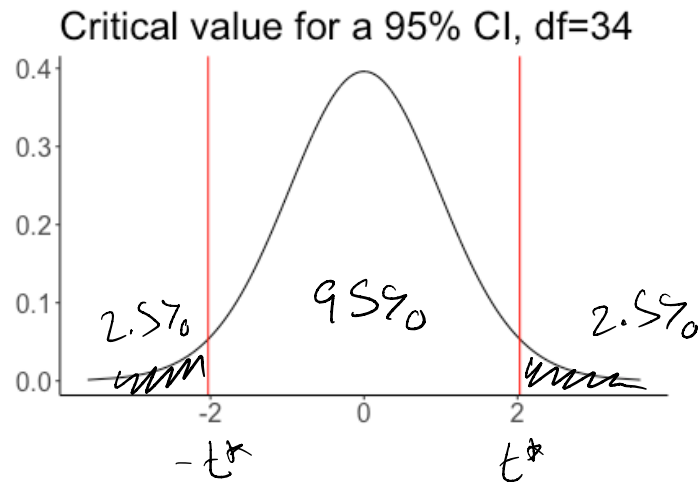
t^* comes from
our desired confidence
level (e.g., 95%)

critical value

(how many standards to go
above & below $\hat{\beta}_1$)

The critical value t^*

To construct a $(1 - \alpha) \cdot 100\%$ confidence interval, t^* is the value such that the area in the right tail of the t_{n-2} distribution is $\alpha/2$

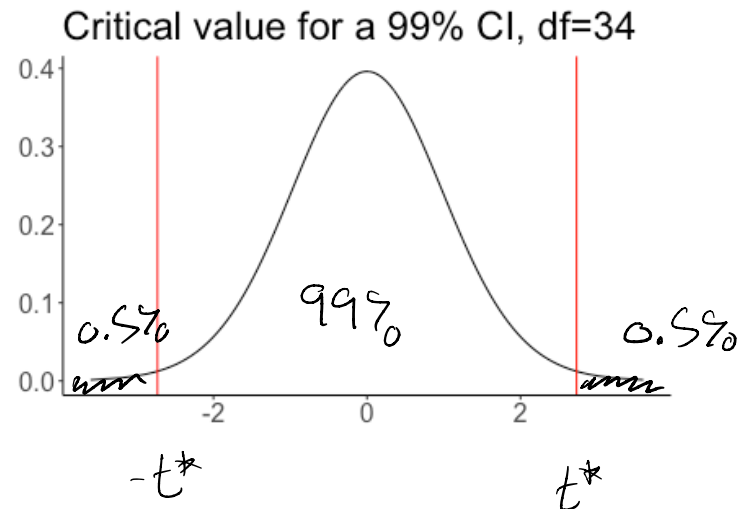


95% CI $\Rightarrow$

$$(1 - 0.05) 100\%$$

$$\Rightarrow \alpha = 0.05$$

$$\Rightarrow \frac{\alpha}{2} = 0.025$$



as confidence level $\uparrow$,
 $t^* \uparrow$

Computing t^*

Example: for a 95% confidence interval and 34 degrees of freedom, $t^* = 2.03$

```
qt(0.025, df=34, lower.tail=FALSE)
```

quantiles of a t distribution
[1] 2.032245

Example: for a 99% confidence interval and 34 degrees of freedom, $t^* = 2.73$

```
qt(0.005, df=34, lower.tail=FALSE)
```

```
## [1] 2.728394
```

Confidence interval

Coefficients:

	Estimate	Std. Error
(Intercept)	87.4277	5.1627
Sugar	<u>2.4808</u>	<u>0.7074</u>

95% confidence interval for β_1 :

$$\hat{\beta}_1 \pm t^* SE\hat{\beta}_1$$

$$t^* = 2.03 \quad (34 \text{ df})$$

$$2.48 \pm 2.03(0.7074) = (1.04, 3.92)$$

Confidence interval

Coefficients:

	Estimate	Std. Error
(Intercept)	87.4277	5.1627
Sugar	2.4808	0.7074

95% confidence interval for β_1 :

$$2.48 \pm 2.03(0.71) = (1.04, 3.92)$$

How do I interpret this confidence interval?

Confidence interval

Coefficients:

	Estimate	Std. Error
(Intercept)	87.4277	5.1627
Sugar	2.4808	0.7074

95% confidence interval for β_1 :

$$2.48 \pm 2.03(0.71) = (1.04, 3.92)$$

Interpretation: We are 95% confident that a one-gram increase in sugar per serving of breakfast cereal is associated with an increase of between 1.04 and 3.92 calories per serving.

What does it mean to be "95% confident"?

if we were to repeat the process and take many samples and calculate many 95% CIs, then 95% of my intervals would contain the β_1

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(..., df = ..., lower.tail=FALSE)
```

What do I fill in to calculate the critical value t^* ?

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(0.01, df = 114, lower.tail=FALSE)
```

```
## [1] 2.359504
```

Activity

	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

$$t^* = 2.36$$

What is my 98% confidence interval?

Activity

	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

$$t^* = 2.36$$

$$1.31 \pm 2.36(0.098) = (1.08, 1.54)$$

Your friend claims that there is a 2% chance that the true slope β_1 is greater than 1.54 or less than 1.08. Is your friend correct?